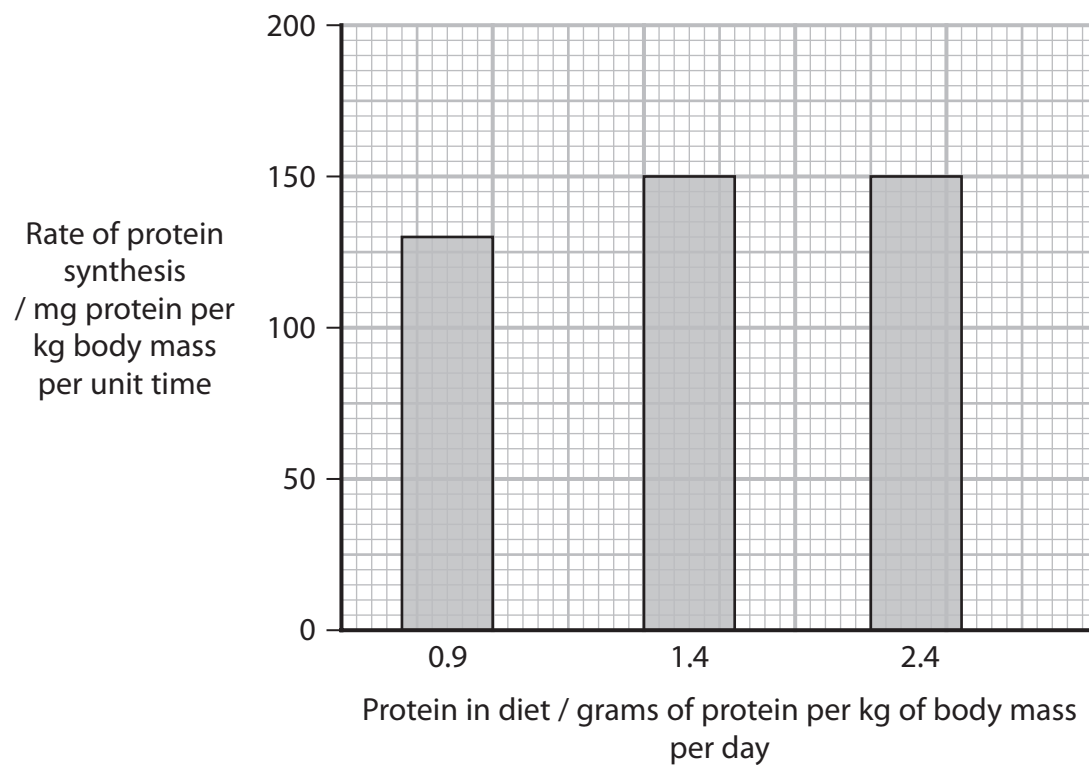


3 Amino acids in the diet of animals are used to synthesise proteins.

(a) The rate of protein synthesis is dependent on a number of factors.

The graph shows the effect of the mass of protein in the diet on the rate of protein synthesis.



Calculate the minimum mass of protein a person with a mass of 70 kg should eat in one week if they want a rate of protein synthesis of 150 mg protein per kg body mass per unit time.

(2)

Answer g

(b) (i) Name the **two** stages in protein synthesis, in the order that they occur.

(1)

Stage 1

Stage 2

(ii) Describe the roles of messenger RNA (mRNA) and transfer RNA (tRNA) in protein synthesis.

(4)

(Total for Question 3 = 7 marks)

